

# FUNCTIONS OF PANCREATIC JUICE

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- **Pancreatic juice has :—**
  - a. Digestive functions
  - b. Neutralizing action

## 1. DIGESTIVE FUNCTIONS

- Digestion of **proteins & lipids**, and mild digestive action on **carbohydrates**.

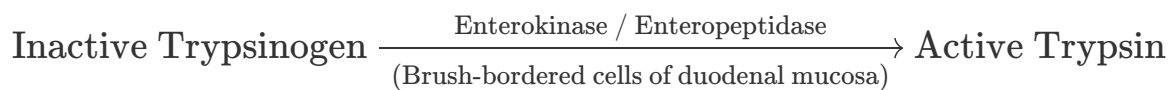
# A. DIGESTION OF PROTEINS

- **Major proteolytic enzymes of pancreatic juice (PJ) :—**

- Trypsin
- Chymotrypsin
- Carboxypeptidase
- Nuclease
- Elastase
- Collagenase

## 1. Trypsin

- Single polypeptide (MW = 25,000), contains 229 amino acids.
- **Activation :—**



↓

Autocatalytic / Autoactive action (itself activates trypsinogen)

- **Trypsin Inhibitor :—**

- If trypsin is activated inside the pancreas → hydrolyzes pancreatic tissue proteins → **pancreatic damage**.
- Premature activation in secretory cells, acini & ducts is prevented by **Trypsin Inhibitor**.

- **Actions of Trypsin :—**

- a. **Digestion of Proteins (Endopeptidase)** : Breaks interior bonds of proteins to convert them into **proteoses & polypeptides**.
- b. **Curdling of Milk** : Converts caseinogen to casein.
- c. **Blood Clotting** : Accelerates blood clotting.
- d. **Activates Other Enzymes of Pancreatic Juice** :
  - Chymotrypsinogen → Chymotrypsin
  - Procarboxypeptidases → Carboxypeptidases
  - Proelastase → Elastase
  - Procolipase → Colipase

- e. **Activates collagenase, phospholipase A & phospholipase B.**
- f. **Autocatalytic action.**

## 2. Chymotrypsin

- Polypeptide (MW = 25,700), contains 246 amino acids.
- Secreted as inactive **chymotrypsinogen**  $\xrightarrow{\text{Trypsin}}$  **Active chymotrypsin.**
- **Actions of Chymotrypsin :—**
  - a. **Digestion of Proteins (Endopeptidase) :** Converts proteins  $\longrightarrow$  polypeptides.
  - b. **Digestion of Milk :** Digests caseinogen faster than trypsin.
  - c. **On Blood Clotting :** No action.

## 3. Carboxypeptidases

- **Types :** Carboxypeptidase A & Carboxypeptidase B (*both activated by Trypsin from precursors Procarboxypeptidase A & B*).
- **Actions (Exopeptidases) :** Break terminal bonds of protein.
  - a. **Carboxypeptidase A :** Converts proteins to amino acids with **aromatic or aliphatic side chains.**
  - b. **Carboxypeptidase B :** Converts proteins to amino acids with **basic side chains.**

## 4. Nucleases

- **Ribonuclease & Deoxyribonuclease :** Digestion of nucleic acids.

RNA & DNA  $\longrightarrow$  Mononucleotides

## 5. Elastase

- Secreted as inactive **proelastase**  $\xrightarrow{\text{Trypsin}}$  **Active elastase.**
- **Action :** Digests elastic fibers.

## 6. Collagenase

- Secreted as inactive **procollagenase**  $\xrightarrow{\text{Trypsin}}$  **Active collagenase**.
- **Action** : Digests collagen.

## B. DIGESTION OF LIPIDS

- **Lipolytic enzymes in pancreatic juice :—**
  - Pancreatic lipase
  - Cholesterol ester hydrolase
  - Phospholipase A
  - Phospholipase B
  - Colipase
  - Bile-salt-activated lipase

### 1. Pancreatic Lipase

- Powerful lipolytic enzyme.
- **Action** : Digests Triglycerides  $\longrightarrow$  Monoglycerides + Fatty Acids.
- **Optimum pH required** : 7 – 9.
- **Digestion of fat requires 2 factors :—**
  - a. **Bile salts** : For emulsification of fat prior to digestion.
  - b. **Colipase** : Coenzyme necessary for pancreatic lipase activity.

#### Steatorrhea :—

Deficiency of this enzyme leads to **excretion of undigested fat in feces**.

### 2. Cholesterol Ester Hydrolase

- Also known as cholesterol esterase.



### 3. Phospholipase A

- Activated by **trypsin**.
- Digests phospholipids:

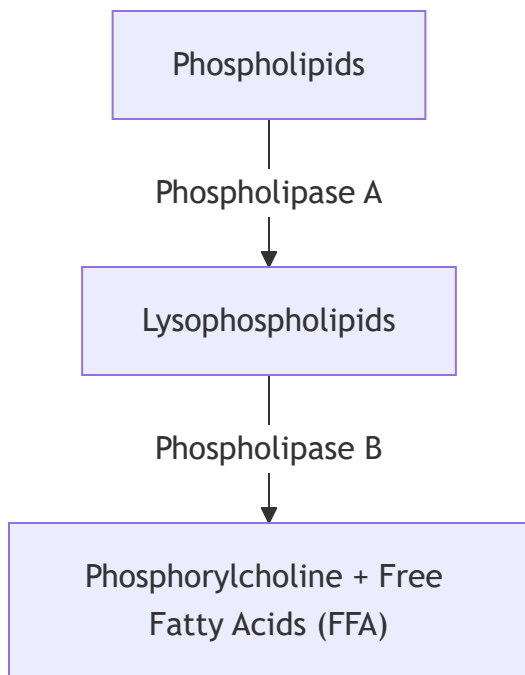
Lecithin  $\longrightarrow$  Lysolecithin

Cephalin  $\longrightarrow$  Lysocephalin

*(These products are lysophospholipids).*

### 4. Phospholipase B

- Activated by **trypsin**.
- Converts lysophospholipids into **phosphorylcholine & free fatty acids (FFA)**.



### 5. Colipase

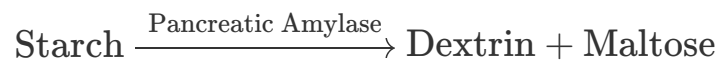
- Secreted as inactive **procolipase**  $\xrightarrow{\text{Trypsin}}$  **Colipase**.
- **Action** : Facilitates the digestive action of pancreatic lipase on fats.

## 6. Bile-Salt-Activated Lipase

- Lipolytic enzyme activated by bile salts (*also called carboxyl ester lipase / cholesterol esterase*).
- **Action** : Hydrolyzes a variety of lipids (phospholipids, cholesterol esters & triglycerides).

## C. DIGESTION OF CARBOHYDRATES

### Pancreatic Amylase (Amylolytic Enzyme)



## 2. NEUTRALIZING ACTION

- **Alkaline pancreatic juice** (*due to presence of bicarbonate ions,  $\text{HCO}_3^-$* ) neutralizes the acidity of chyme entering the intestine.
- **Importance** : Protects the intestinal mucosa from the destructive erosive action of acid chyme.

**TABLE 39.1: Digestive enzymes of pancreatic juice**

| Enzyme                      | Activator               | Acts on (substrate)                     | End products                            |
|-----------------------------|-------------------------|-----------------------------------------|-----------------------------------------|
| Trypsin                     | Enterokinase<br>Trypsin | Proteins                                | Proteoses and polypeptides              |
| Chymotrypsin                | Trypsin                 | Proteins                                | Polypeptides                            |
| Carboxypeptidases           | Trypsin                 | Polypeptides                            | Amino acids                             |
| Nucleases                   | Trypsin                 | RNA and DNA                             | Mononucleotides                         |
| Elastase                    | Trypsin                 | Elastin                                 | Amino acids                             |
| Collagenase                 | Trypsin                 | Collagen                                | Amino acids                             |
| Pancreatic lipase           | Alkaline medium         | Triglycerides                           | Monoglycerides and fatty acids          |
| Cholesterol ester hydrolase | Alkaline medium         | Cholesterol ester                       | Cholesterol and fatty acids             |
| Phospholipase A             | Trypsin                 | Phospholipids                           | Lysophospholipids                       |
| Phospholipase B             | Trypsin                 | Lysophospholipids                       | Phosphoryl choline and free fatty acids |
| Colipase                    | Trypsin                 | Facilitates action of pancreatic lipase | –                                       |
| Bile-salt-activated lipase  | Trypsin                 | Phospholipids                           | Lysophospholipids                       |
|                             |                         | Cholesterol esters                      | Cholesterol and fatty acids             |
|                             |                         | Triglycerides                           | Monoglycerides and fatty acids          |
| Pancreatic amylase          | –                       | Starch                                  | Dextrin and maltose                     |